The documentation and process conversion measures necessary to comply with this amendment shall be completed by 2 July 1998.

IINCH-POUND

MIL-PRF-19500/592C
AMENDMENT 1
2 July 1998

PERFORMANCE SPECIFICATION

SEMICONDUCTOR DEVICE, REPETITIVE AVALANCHE, FIELD EFFECT TRANSISTOR, N-CHANNEL, SILICON, TYPES 2N7224, 2N7225, 2N7227, 2N7228, 2N7224U, 2N7225U, 2N7227U, AND 2N7228U JANTX, JANTXV, JANS, JANHC AND JANKC

This amendment forms a part of MIL-PRF-19500/592C, dated 21 June 1996, and is approved for use by all Departments and Agencies of the Department of Defense.

The attached insertable replacement pages listed below are replacements for stipulated pages. When the new pages have been entered in the document, insert the amendment as the cover sheet to the specification.

Replacement page	Pages Replaced
5	5
6	6
11	11
12	12

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Navy - TD
Air Force - 70, 80

Preparing activity:
DLA - CC

(Project 5961-2043)

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited,

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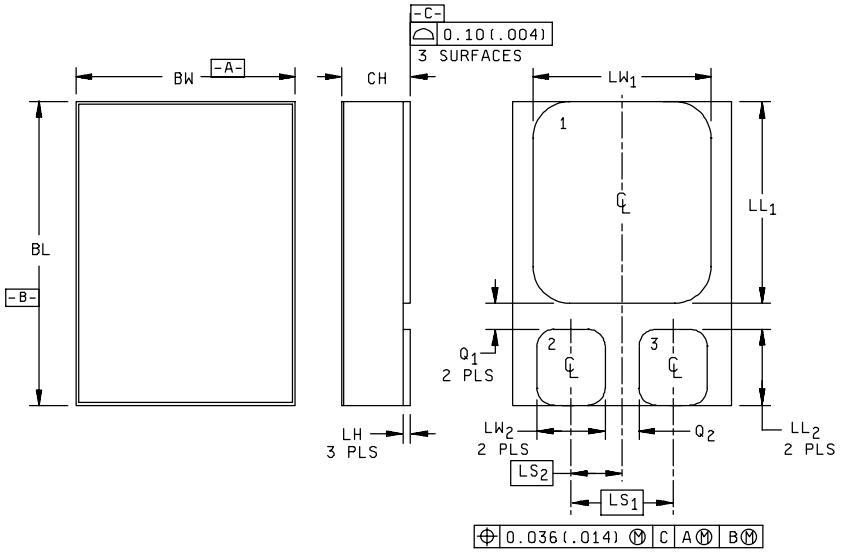

Dimensions				
Symbol	Inches		Millimeters	
	Min	Max	Min	Max
BL	.619	.631	15.73	16.02
BW	.444	.456	11.28	11.58
CH	- - -	.142	- - -	3.60
LH	.010	.020	0.26	0.50
LL_1	.409	.421	10.39	10.69
LL_2	.151	.163	3.84	4.14
LS_1	.210 BSC		5.33 BSC	
LS_2	.105 BSC		2.67 BSC	
LW_1	.369	.381	9.38	9.67
LW_2	.134	.146	3.41	3.70
Q_1	.030	- - -	0.76	- - -
Q_2	.035	- - -	0.89	- - -
Term 1	Drain			
Term 2	Gate			
Term 3	Source			

NOTES:
1. Controlling dimension: Inch.
2. Metric equivalents are given for information only.
3. The lid shall be electrically isolated from the drain, gate and source.
4. Dimensioning and Tolerancing shall be in accordance with ANSI Y14.5M 1982.

FIGURE 2. Dimensions and configuration of surface mount package outline .

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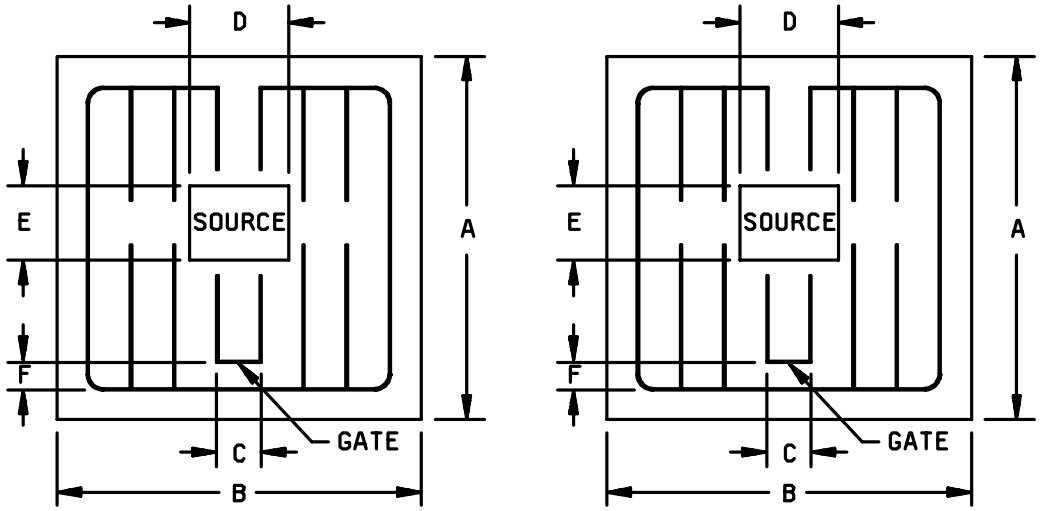
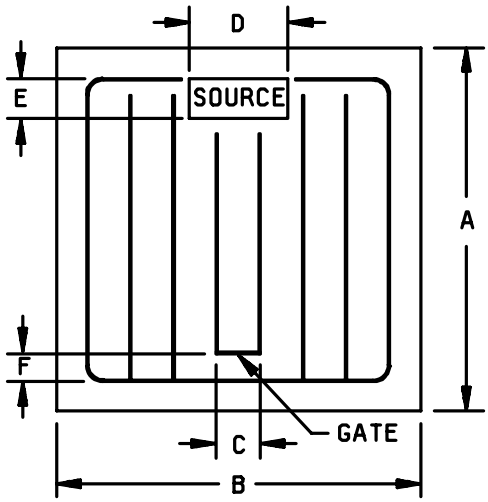


2N7224

2N7225



2N7227 AND 2N7228

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerance is ±0.005 inches (0.13 mm).
4. Physical characteristics of the die thickness = .0187 inch (0.47 mm).
5. Back metal: Cr - Ni - Ag.
6. Top metal: Al.
7. Back contact: Drain.

FIGURE 3. Physical dimensions JANHC and JANKC.

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4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VIa (JANS) and table VIb (JANTX and JANTXV) of MIL-PRF-19500 and as follows. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table I, subgroup 2 herein.

4.4.2.1 Group B inspection, table VIa (JANS) of MIL-PRF-19500.

Subgroup	Method	Conditions
B3	1051	Test condition G.
B3	2037	Test condition A. All internal wires for each device shall be pulled separately. If group B3 is to be continued to C6, strength test may be performed after C6.
B4	1042	Test condition D, 2,000 cycles. The heating cycle shall be 1 minute minimum. No heat sink nor forced air cooling on the device shall be permitted during the "on" cycle.
B5	1042	A separate sample may be pulled for each test. Accelerated steady-state reverse bias; test condition A, V_{DS} = rated, T_A = +175° C, t = 120 hours, read and record $V_{BR(DSS)}$ (pre and post) at I_D = 1 mA. Read and record I_{DSS} (pre and post) in accordance with table I, subgroup 2 herein. $V_{BR(DSS)}$ delta cannot exceed 10 percent.
B5	1042	Accelerated steady-state gate stress; test condition B, V_{GS} = rated, T_A = +175° C, t = 24 hours.
B6		See 4.5.2.

4.4.2.2 Group B inspection, table VIb (JANTX and JANTXV) of MIL-PRF-19500.

Subgroup	Method	Conditions
B2	1051	Test condition G.
B3	1042	Test condition D, 2,000 cycles. The heating cycle shall be 1 minute minimum. No heat sink nor forced air cooling on the device shall be permitted during the "on" cycle.
B3	2037	Test condition A. All internal wires for each device shall be pulled separately. If group B3 is to be continued to C6, bond strength test may be performed after C6.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500, and as follows. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps of table I, subgroup 2 herein.

Subgroup	Method	Conditions
C2	2036	Test condition A; weight = 10 pounds, t = 15 s (not applicable for surface mount devices).
C6	1042	Test condition D, 6,000 cycles. The heating cycle shall be 1 minute minimum. No heat sink nor forced air cooling on the device shall be permitted during the "on" cycle.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

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4.5.2 Thermal resistance. Thermal resistance measurements shall be performed in accordance with method 3161 of MIL-STD-750. $R_{\theta JC(max)}$ = .83°C/W for TO-254AA case style devices and surface mount devices. The following parameter measurements shall apply:

a. I_M measuring current ..... 10 mA.

b. I_H drain heating current..... 3 A.(5A minimum for surface mount devices).

c. t_H heating time ..... Steady state (see MIL-STD-750, method 3161).

d. V_H drain-source heating voltage..... 25 V minimum (20V minimum for surface mount devices).

e. t_{MD} measurement time delay..... 30 to 60 µs.

f. t_{SW} sample window time..... 10 µs maximum.

4.5.3 Thermal impedance ($Z_{\theta JC}$ measurements). The $Z_{\theta JC}$ measurements shall be performed in accordance with method 3161 of MIL-STD-750. The maximum limit (not to exceed the group A, subgroup 2 limit or figure 4 thermal impedance curve) for $Z_{\theta JC}$ in screening (table IV of MIL-S-19500) shall be derived by each vendor by means of statistical process control. When the process has exhibited control and capability, the capability data shall be used to establish the fixed screening limit. In addition to screening, once a fixed limit has been established, monitor all future sealing lots using a random five piece sample from each lot to be plotted on the applicable X, R chart. If a lot exhibits an out of control condition, the entire lot shall be removed from the line and held for Engineering evaluation and disposition. This procedure may be used in lieu of an in line monitor. The following parameter measurements shall apply:

a. I_M measuring current ..... 10 mA.

b. I_H drain heating current ..... 3 A minimum (5A minimum for surface mount devices).

c. t_H heating time ..... 100 ms minimum (30 ms minimum for surface mount devices).

d. V_H drain-source heating voltage ..... 25 V minimum (20 V minimum for surface mount devices).

e. t_{MD} measurement time delay..... 30 to 60 µs.

f. t_{SW} sample window time..... 10 µs maximum.

4.5.4 Single pulsed unclamped inductive switching.

a. Peak current, I_D ..... I_{AR}(max)

b. Peak gate voltage, V_{GS} ..... 10 V.

c. Gate to source resistor, R_{GS}..... $25 \leq R_g \leq 200\ \Omega$.

d. Initial case temperature ..... +25°C, +10°C, -5°C.

e. Inductance, L ..... $\left[\frac{2E_{AS}}{(I_{D1})^2}\right]\left[\frac{(V_{BR}-V_{DD})}{V_{BR}}\right]$ mH minimum.

f. Number of pulses to be applied..... 1 pulse minimum.

g. Supply voltage (V_{DD})..... 50 V, (25 V for devices with minimum $V_{(BR)DSS}$ of 100 V).

4.5.5 Gate stress test..... V_{GS} = ±30 V minimum; t = 250 µs minimum.

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